curriculum vitae

Computational Materials Engineer with a Ph.D. in Computational & Experimental Mechanics specializing in physics-based surrogate modeling, multi-scale simulation, large-scale uncertainty quantification (UQ) and experimental-numerical methods. Proven ability to bridge high-fidelity physics simulations with practical engineering predictions through large-scale data curation, custom Python tooling, and production-grade surrogate models. Strong background in data-driven reduced-order modeling, uncertainty quantification, and deploying scalable multi-physics pipelines in HPC environments. Passionate about using physics-informed and data-driven methods to democratize advanced simulation and enable digital twin applications.

Personalia

Surname:	Ruybalid
Given names	Andre, Paul
Degrees:	Ph.D. | MSc. | BSc.
City:	Los Alamos, New Mexico, U.S.A.
Phone number:	+1 719 210 1746 (personal mobile)
E-mail:	andreruybalid@gmail.com
Date of birth:	26 October 1987
Place of Birth:	Farmington, New Mexico, U.S.A.
Nationality:	American/Dutch

Professional experience

2023 - Present **R&D Engineer, Los Alamos National Laboratory**

Materials Science and Technology Division (MST-8) | Los Alamos, NM, U.S.A.

- Architected and deployed "LAROMance", a physics-informed surrogate modeling framework for viscoplastic creep in nuclear reactor materials, enabling 1000x+ speedup over full-order simulations while preserving mechanistic accuracy. Two-time R&D 100 Award Finalist.
- Designed large-scale simulation pipelines that curated datasets from tens of thousands of low-length-scale simulations, applying ablation studies and dimensionality reduction to optimize model fidelity vs. computational cost.
- Developed and integrated surrogate constitutive models into the MOOSE/BISON Finite Element (FEA) framework, including Verification & Validation (V&V), and large-scale uncertainty quantification (UQ).
- Built modular Python tooling for guided workflow orchestration, data processing, and visualization, accelerating analysis cycles and enabling reproducible, production-grade deployments of physics-based surrogates.

2021 – 2023 **Postdoctoral Researcher, Los Alamos National Laboratory**

Materials Science and Technology Division (MST-8) | Los Alamos, NM, U.S.A.

- Developed a novel surrogate modeling methodology specifically for history-dependent material behavior, using regression methods on data generated from spectral (FFT) polycrystal simulations.
- Curated and managed large-scale simulation datasets from thousands of high-fidelity runs; designed and implemented preprocessing, feature engineering, and surrogate model calibration pipelines.

2019 – 2020 **Design Engineer, ASML**

DUV Def & Metrology group of Defectivity Department | Eindhoven, The Netherlands

Developed and maintained complex numerical algorithms and custom MATLAB tooling for high-precision nanoscale defect detection and geometry analysis on DUV lithography reticles. Designed, validated, and optimized metrology and data processing pipelines through large-scale data analysis and cleanroom experimentation, delivering production-ready solutions under tight deadlines in cross-functional industrial environments.

2013 – 2019 **PhD Researcher, Eindhoven University of Technology**
Computational and Experimental Mechanics | Prof. Dr. Marc Geers & Dr. Johan Hoefnagels.
Philips Research, Materials Innovation Institute | Eindhoven, The Netherlands

- Designed and implemented an integrated experimental-numerical framework in MATLAB, combining Finite Element Modeling in MSC Marc, image processing, and Digital Image Correlation (IDIC) for full-field material characterization and inverse parameter identification.
- Developed virtual experimentation frameworks by applying known displacement fields to deform experimental images, then tested new Digital Image Correlation (IDIC) algorithms through inverse identification and sensitivity analysis.

Education

Eindhoven University of Technology, the Netherlands
Department of Mechanical Engineering
Division: Computational and Experimental Mechanics
Research Group: Mechanics of Materials

2013 – 2019 **PhD. in Mechanical Engineering**
Title: "Identification of multi-layer interface systems: a full-field, micro-mechanical approach."

2013 – 2019 **Graduate School of Engineering Mechanics**
Advanced Mechanics Courses

2010 – 2013 **MSc. Mechanical Engineering**
Minor: Biomedical Engineering

2006 – 2010 **BSc. Mechanical Engineering**

2000 – 2006 **Research-oriented secondary education (VWO)**
St.-Janscollege, Hoensbroek, The Netherlands

Technical Skills

Programming & Software Tools	Python, MATLAB, C++, Linux/bash scripting, SLURM (HPC), MSC Marc/Mentat, Paraview, LLM-assisted development (Continue.dev in VS Code)
Scientific Computing	Physics-based surrogate modeling, reduced-order modeling, spectral methods (FFT), finite element method (FEM), sparse linear solvers and regression, large-scale dataset curation & ablation studies, inverse optimization, uncertainty quantification, feature engineering
Software Development	Custom CAE tool & pipeline development, workflow automation, interactive Python tooling (GUI), Python-to-C++ integration, production-grade code development, modular design, Git
Domain Expertise	Computational mechanics, multi-scale & multi-physics modeling, continuum mechanics, viscoplasticity and creep modeling, cohesive zone and interface mechanics, plasticity and dislocation mechanics
Technical Approaches	In-situ small-scale mechanical testing, high-resolution microscopy (SEM, SPM, optical profilometry), Digital Image Correlation (IDIC), experimental-numerical frameworks

Selected journal publications (peer-reviewed)

P. Robbe, A.P. Ruybalid, A. Hegde, C. Bonneville, H.N. Najm, L. Capolungo and C. Safta . A comparison of surrogate constitutive models for viscoplastic creep simulation of HT-9 steel. *Computational Materials Science*, 262, 114367 (2026). https://doi.org/10.1016/j.commatsci.2025.114367.

A.P. Ruybalid, A. Tallman, W. Wen, C. Matthews, L. Capolungo. Data-Driven Surrogate Modeling with Microstructure-Sensitivity of Viscoplastic Creep in Grade 91 Steel. Integrating Materials and Manufacturing Innovation 13, 895–914 (2024). https://doi.org/10.1007/s40192-024-00377-z

A.P. Ruybalid, O. van der Sluis, M.G.D. Geers and J.P.M. Hoefnagels. Full-field identification of mixed-mode adhesion properties in a flexible, multi-layer microelectronic material system. *Engineering Fracture Mechanics*, 226, pp. 106879 (2020), *https://doi.org/10.1016/j.engfracmech.2020.106879*.

A.P. Ruybalid, J.P.M. Hoefnagels, O. van der Sluis, M.P.F.H.L. van Maris and M.G.D. Geers. Mixed-mode cohesive zone parameters from integrated digital image correlation on micrographs only. *International Journal of Solids and Structures*, 156-157, pp. 179-193 (2019), *https://doi.org/10.1016/j.ijsolstr.2018.08.010*.